



Free and Open Source Tools for Research

Jesse Oldroyd

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Free and Open Source Tools for Research

- What is FOSS?

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Definition

Free and open source software (FOSS) is software

- that is freely available (usually under a permissive license);

- has accessible source code.

Free vs. Open

More restrictive definitions exist [1]!

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Philosophy of Free Software

The philosophy of free software can be encapsulated by *four essential freedoms* [2]:

1. to use the software.
2. to inspect and change the software.
3. to share copies of the original software.
4. to share copies of the modified software.

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The freedoms associated with the use of free software are given legal weight by various **licenses**, classified as *copyleft* or *permissive* [3].

- Copyleft licenses: protects against proprietary software development.

- Permissive: encourages adoption and collaboration.

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└ What is FOSS?

└ Licenses

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When FOSS Is the Right Choice

FOSS might be the right choice if you require

- Transparency.
- Composability.
- Customizability.

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Free and Open Source Tools for Research

└ What is FOSS?

└ When FOSS Is the Right Choice

1. Transparency lets you inspect software itself to verify results. Code that uses proprietary software makes it difficult, if not impossible, to confirm the correctness of results or to identify errors.
2. Free software is often text-based. Since so many use this common format, it can be relatively painless to chain multiple tools together into a single workflow.
3. Given how permissive many open source tools are, it's possible (and often expected) for you to customize the tool to your own needs.

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When FOSS Is the Wrong Choice

Despite good reasons to use them, free and open source tools are not always a good idea. You need to be mindful of the following:

- Use tools that work well with what your collaborators are using.
- FOSS tools are often not as polished as their proprietary counterparts.
- Security concerns.

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└ What is FOSS?

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1. In particular, follow any relevant journal guidelines!
2. FOSS tools are also more subject to becoming “abandonware” as maintaining them is time-consuming—and often thankless—work.
3. While many open-source projects are well-maintained, their accessibility to the public does leave them at risk of being used by malicious actors. You should only use tools from trustworthy sources!

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Free and Open Source Tools for Research
└─ FOSS Tools

Tools

Tools

Types of Tools

- There is a large selection of open source tools available, including **coding** and **writing**.
- One of the most fundamental tools is a text editor.
- Even though you start with text, output is not necessarily just a text file!

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- R
- Python with pandas, Seaborn and SciPy
- Python with NumPy
- Python with SymPy

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Free and Open Source Tools for Research

└─ FOSS Tools

└─ Coding

└─ Tools for Computation

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Python Code for Statistics

```
1 import numpy as np
2 import seaborn as sns
3 import matplotlib.pyplot as plt
4 sns.set_theme(style="dark")
5
6 # Simulate data from a bivariate Gaussian
7 n = 10000
8 mean = [0, 0]
9 cov = [(2, .4), (.4, .2)]
10 rng = np.random.RandomState(0)
11 x, y = rng.multivariate_normal(mean, cov, n).T
12
13 # Draw a combo histogram and scatterplot with density
    contours
14 f, ax = plt.subplots(figsize=(6, 6))
15 sns.scatterplot(x=x, y=y, s=5, color=".15")
16 sns.histplot(x=x, y=y, bins=50, pthresh=.1, cmap="mako")
17 sns.kdeplot(x=x, y=y, levels=5, color="w", linewidths=1)
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Listing 1: Example Python code for statistical visualization

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Figure 1: Data visualization produced by Listing 1

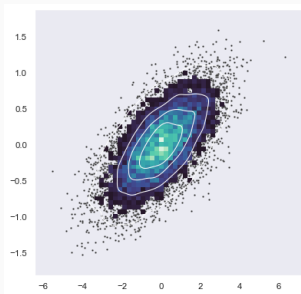


Figure 1: Data visualization produced by Listing 1

- When using programming languages, you will need a program to interact with the language.
- There is often a choice between an *integrated development environment* (IDE) and a text editor.
- Open source text editors: VSCode (new school), Vi(m) and Emacs (very old school).

General or specific language support

IDEs are tailored to specific languages (such as RStudio). Text editors tend to be more multi-purpose.

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- L^AT_EX

- git

- harper-ls and LanguageTool

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└─ FOSS Tools

└─ Writing

└─ Tools for Writing

1. Perhaps the most popular open-source tool for creating technical documents out there. Primarily used in mathematics and physics, but there are also packages available for chemistry, biology and more.
2. A tool for managing different versions of a document. This is useful for tracking changes in any text-based document.
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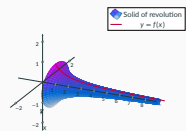


Figure 2: A PGFPlots drawing

- matplotlib
- pandas
- TikZ/PGFPlots
- Inkscape

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FOSS Tools

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Tools for Drawing

Tools for Drawing



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- The tools mentioned so far are not mutually exclusive!

- These tools have specific purposes, but can be chained together to form a powerful workflow.

Solving a Math Problem

The expanded form of $(x - 3)^5$ is ...

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- GitHub
- personal website.

Dangers of Sharing Work

You should be careful when sharing potentially sensitive work.
Always check with your advisor and collaborators!

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Beware of AI training!

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- Work or results that depend on computer analysis can be shared using Jupyter Notebooks.

- Sharing code and a narrative is often more instructive than just sharing the code itself. See: [gravity wave notebooks](#).

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└ FOSS Tools

└ Publicizing

└ Literate Programming

1. Other tools include Quarto and Typst.
2. Notebooks can also be hosted via Google Colab for interactivity.

- Work or results that depend on computer analysis can be shared using Jupyter Notebooks.

- Sharing code and a narrative is often more instructive than just sharing the code itself. See: [gravity wave notebooks](#).

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- Despite their utility, these tools often take longer to learn and are not as immediately intuitive as their proprietary counterparts.
- You need to decide if they are appropriate for your project!

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└ Presentation Repository

